

Project Proposal: The Kinematics of Learning Mapping and Intervening in the Geometric Stabilization of Transformer Matrices

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1 Project Description

Problem & Motivation: The transformer architecture is composed of several weight matrices: embeddings (W_E), attention routing (W_Q, W_K, W_V), and feed-forward memory (W_{FFNN}). Current literature suggests that neural networks stabilize in a bottom-up manner during training (block 1, then block 2 and so on), but it remains unclear *when* the stabilization occurs within blocks. We wish to study the stabilization behavior of the Attention and FFNN matrices within these individual blocks, and build upon this to find ways to artificially control the training phase (using dynamic freezing, learning rates, etc).

What is Known & What We Hope to Achieve: Existing optimization techniques, such as Layer-wise Learning Rate Decay (ULMFiT), control learning rates at the macro-layer level. We aim to achieve a more granular, matrix-level intervention. First, we will map the stabilization timeline of a model during pre-training by calculating the Frobenius norm and cosine similarity of specific weight matrices across intermediate checkpoints. Second, guided by this kinematic mapping, we will utilize dynamic freezing and dynamic learning rates to control the stability and computational emphasis of specific matrices during the training phase. Using PyTorch parameter groups, we will experiment with several targeted learning schedules. These will possibly include an “attention-first” approach (solidifying routing logic before memory), an “attention-second” approach, and periodic alterations where the Attention and FFNN matrices are incrementally emphasized in alternating steps. By allowing these distinct functional mechanisms to feed off each other’s development in isolated, periodic phases, we will test if uncoupling their optimization yields faster optimization than simultaneous global updates.

Furthermore, by observing the network’s stabilization behavior, we will explore combining dynamic freezing with targeted learning rate tuning. This involves

temporarily freezing the stabilized components while changing the learning rate of an isolated matrix, forcing it to dynamically alter its geometric trajectory. This will allow us to observe how granular learning rate manipulations affect the stabilization process and optimize overall performance, without destroying the learned representations of the rest of the network.

All of these suggestions are of course dependent on our findings from the initial stage of studying the weight’s dynamic behavior.

Evaluation: We will evaluate our approach against an identical Transformer trained conventionally (e.g., AdamW, global learning rate, no decoupled matrix interventions)—to precisely isolate the efficiency gains of our targeted schedules. To assess improvements in model capability, we will track validation loss, perplexity, and the emergence of downstream capabilities as a function of training steps. Additionally, we will quantify optimization and computational costs by measuring overall FLOPs and VRAM usage throughout the training process.

2 Assumptions and Requirements

Compute & Hardware: Initial tensor extraction, matrix velocity calculations, and rapid prototyping of the PyTorch parameter group interventions will be conducted locally utilizing an NVIDIA RTX 5070 Ti and AMD Ryzen 7800X3D setup to ensure efficiency. Once a working prototype is completed locally, we may opt to use the GPU cluster of the Tel-Aviv Data Center to obtain more robust results.

Data & Models: The analytical mapping phase of this project requires access to intermediate training checkpoints and sequence-level metadata. We will utilize the LMEnt suite to execute these experiments, as it provides a controlled framework for analyzing capabilities and representations from pretraining data.

Additional Resources: This project relies entirely on open-source suites and structural matrix manipulation. We do not require any external API credits (e.g., OpenAI/Gemini), specialized cloud endpoints, or human evaluators.

References

- [1] Raghu et al. (2017). *SVCCA: Singular Vector Canonical Correlation Analysis for Deep Learning Dynamics*. Provides the mathematical framework for tracking when neural layers geometrically solidify, and using dynamic freezing (“Freeze Training”) to optimize computation and improve generalization.
- [2] Meng et al. (2022). *Locating and Editing Factual Associations in GPT (ROME)*. Proves the functional division of routing (Attention) and memory (FFNN) within Transformer blocks, supporting our goal of trying to decouple the two to improve performance.

- [3] Howard & Ruder (2018). *Universal Language Model Fine-tuning for Text Classification (ULMFiT)*. Introduces layer-wise learning rate decay, serving as our macro-level optimization baseline.